

3. A rotor wake in hover is characterised by the vortices that are shed from the tips of the blade and form a helical structure. Consider a single vortex filament, discretised in a straight segment approximation as shown in Figure 3, for a full revolution (360 degrees). The filaments correspond to an angular step $\Delta\zeta$. Assume that the wake does not contract and therefore the vortex filament forms a perfect helix under the rotor, and that there is no vortex diffusion. The vortex strength remains the same, Γ_v . Also assume that after a revolution, the filament has descended a distance of $0.2R$ in the negative z -direction, where R is the rotor radius.

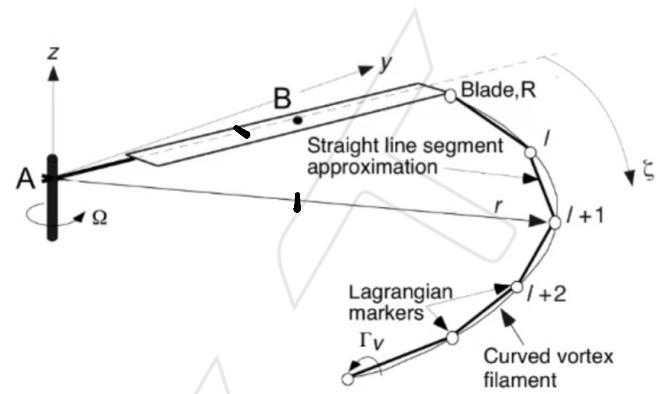
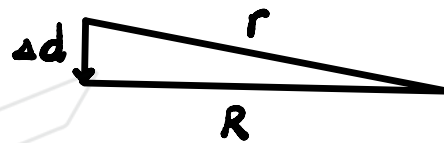
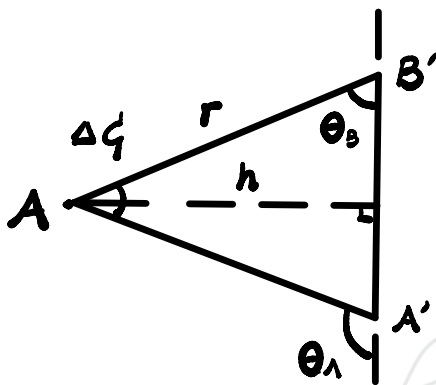


Figure 3: Wake vortex filament discretisation.

$$(a) \quad v_i = \frac{\Gamma}{4\pi h} (\cos\theta_{B'} - \cos\theta_{A'})$$



Therefore, as θ_A , θ_B and R is constant

$$r = \sqrt{R^2 + \Delta d^2} = \sqrt{R^2 + \left[(n-1) \frac{\Delta\zeta}{2\pi} (0.2R)\right]^2}$$

$$\theta_{B'} = \frac{\pi - \Delta\zeta}{2}$$

$$\theta_{A'} = \pi - \frac{\pi - \Delta\zeta}{2} = \frac{\pi + \Delta\zeta}{2}$$

$$h = r \cos\left(\frac{1}{2}\Delta\zeta\right)$$

$$\therefore (v_i)_A = \frac{\Gamma}{4\pi \sqrt{R^2 + \left[\frac{\Delta\zeta}{2\pi} (n-1) 0.2R\right]^2} \cos\left(\frac{1}{2}\Delta\zeta\right)} \left[\cos\left(\frac{\pi - \Delta\zeta}{2}\right) + \cos\left(\frac{\pi + \Delta\zeta}{2}\right) \right]$$

b) $(V_i)_{A, \text{total}} = \sum_{n=1}^{\frac{\pi}{\Delta q}} (V_i)_A$

c) $\Delta q \uparrow \longrightarrow \text{more segment} \longrightarrow \text{larger sum of lengths of the filaments } (\sum A'B') \longrightarrow \text{higher } (V_i)_{A, \text{total}}$

d) higher $\longrightarrow \sum h$ smaller